Spacecraft Attitude Determination and Control

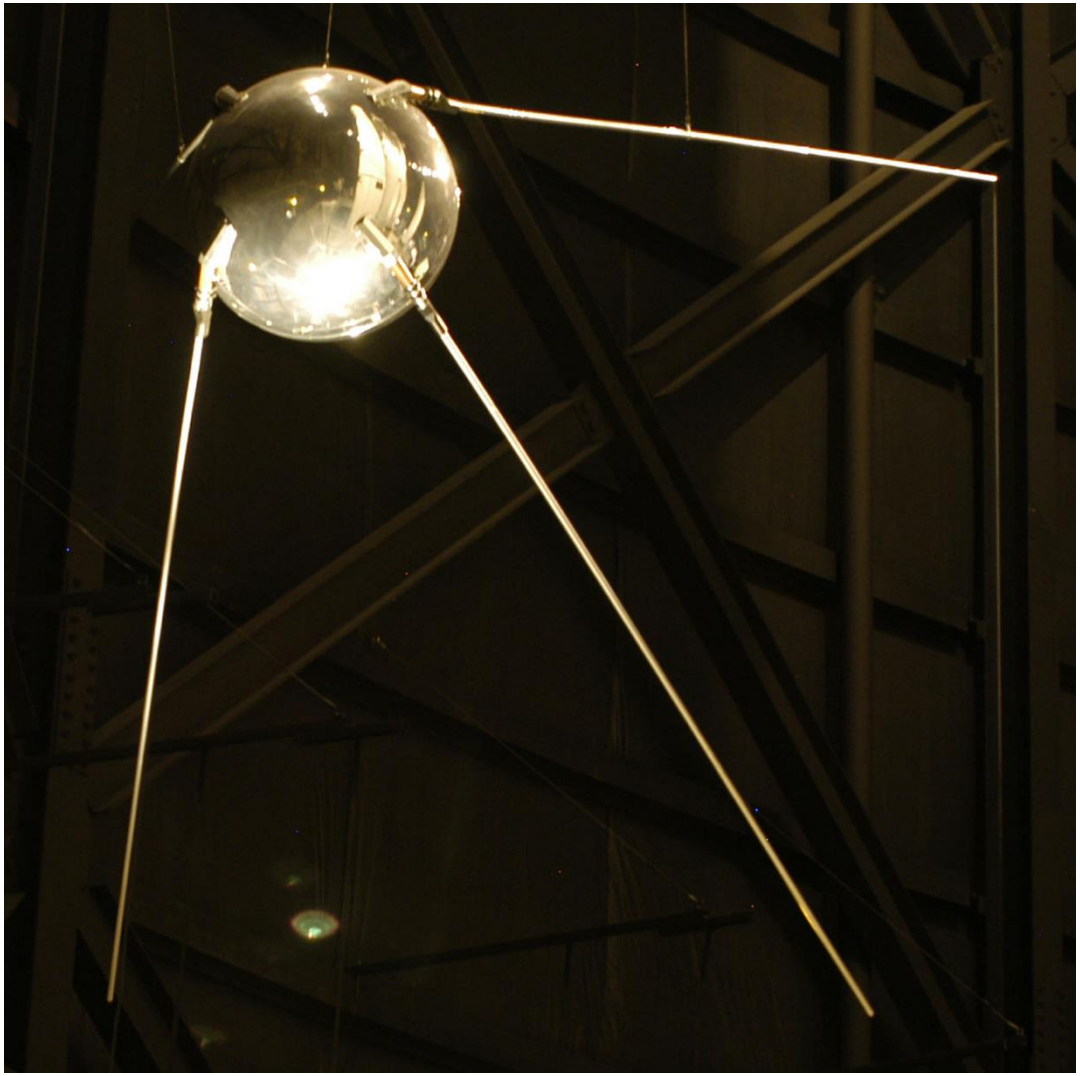
Spring 2026

Zac Manchester

Why We Care About This Stuff

- Attitude determination and control is one of the core disciplines within spacecraft engineering.
- Almost all spacecraft have to address the following questions:
  - What needs to point where?
  - How well does it need to hold attitude?
- Attitude is intimately tied to all other spacecraft functions:
  - Power
  - Thermal
  - Propulsion
  - Telemetry & Command
  - Payload

A Brief History of Attitude…

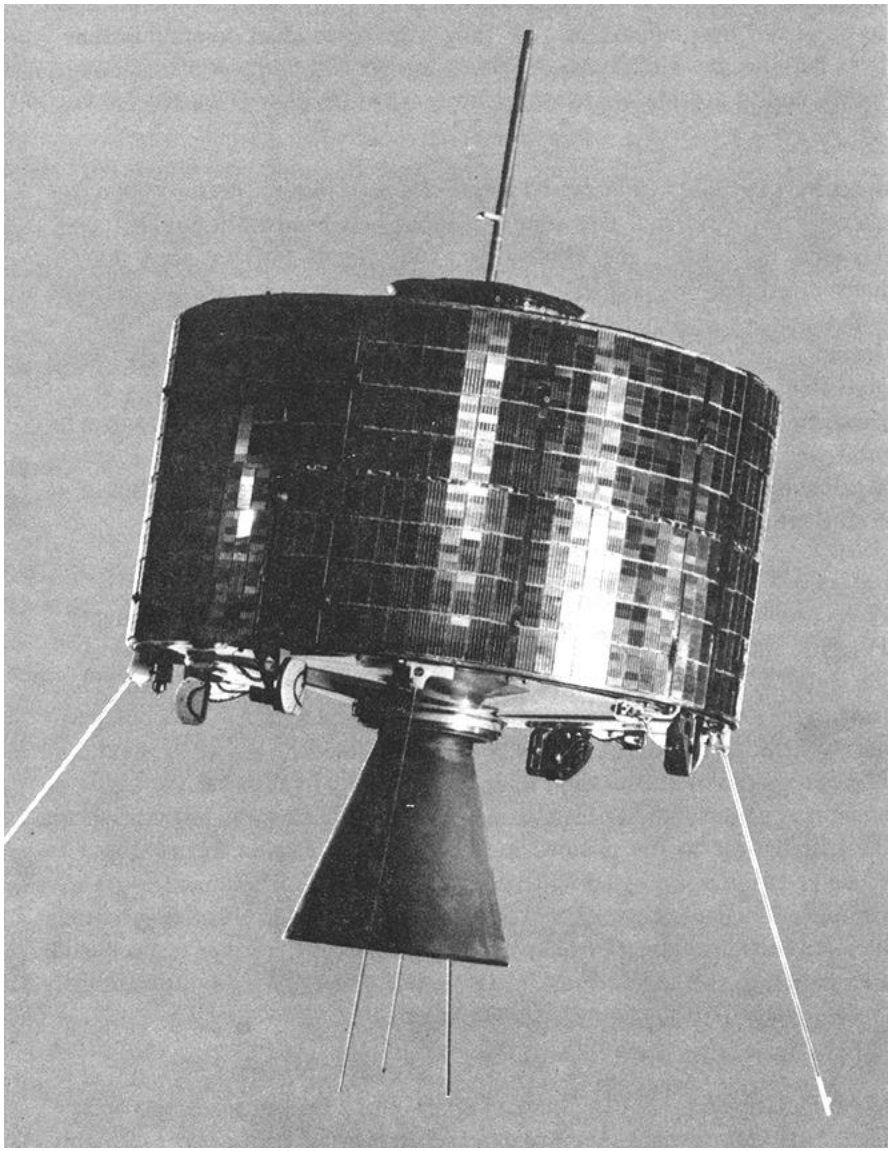


Sputnik - 1957

A Brief History of Attitude…

Explorer 1 - 1958

A Brief History of Attitude…

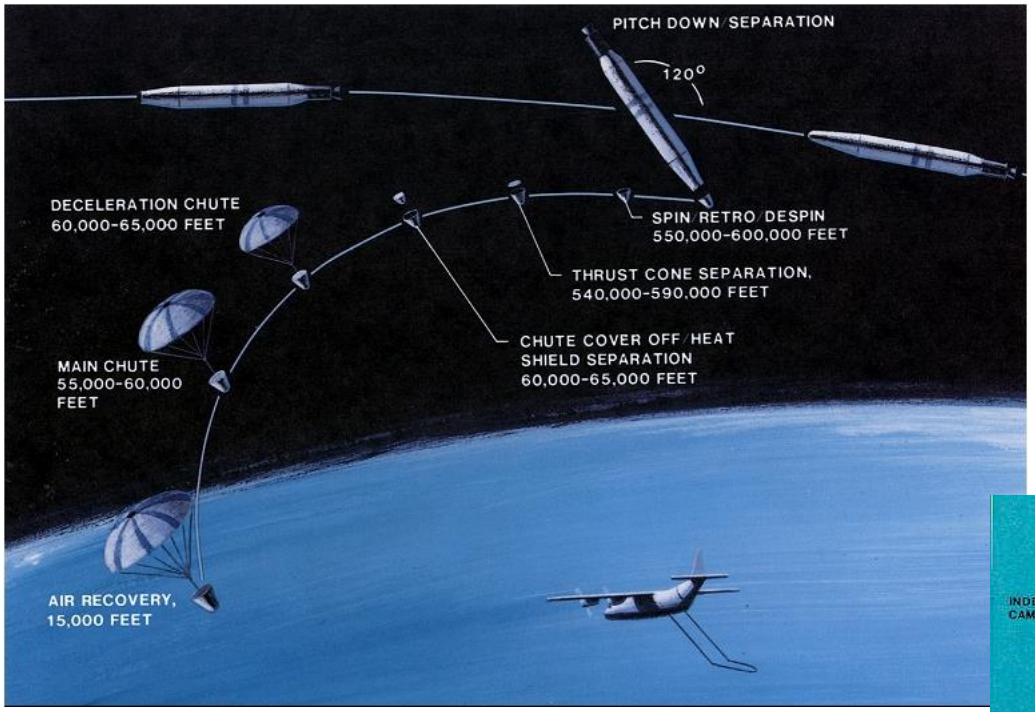


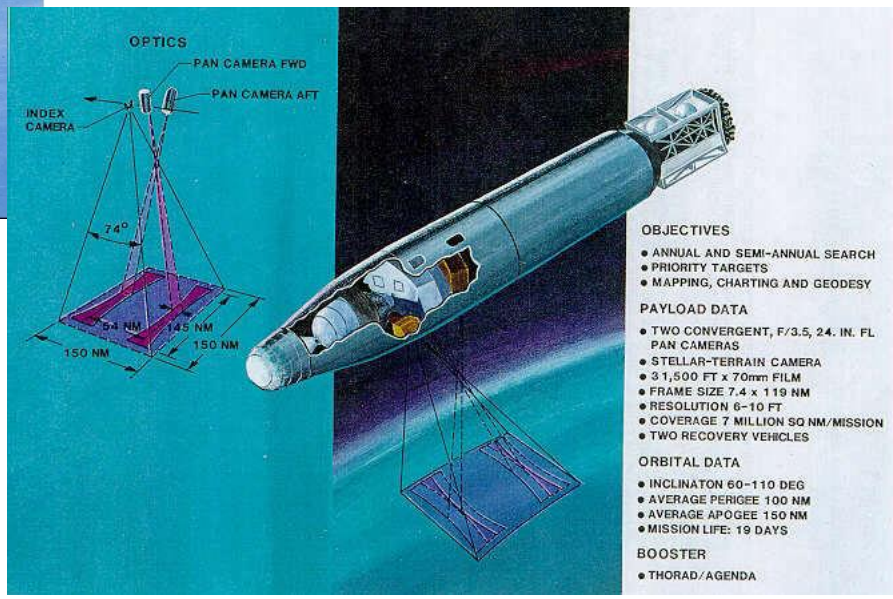
Syncom - 1963

A Brief History of Attitude…

TACSAT 1 - 1969

A Brief History of Attitude…



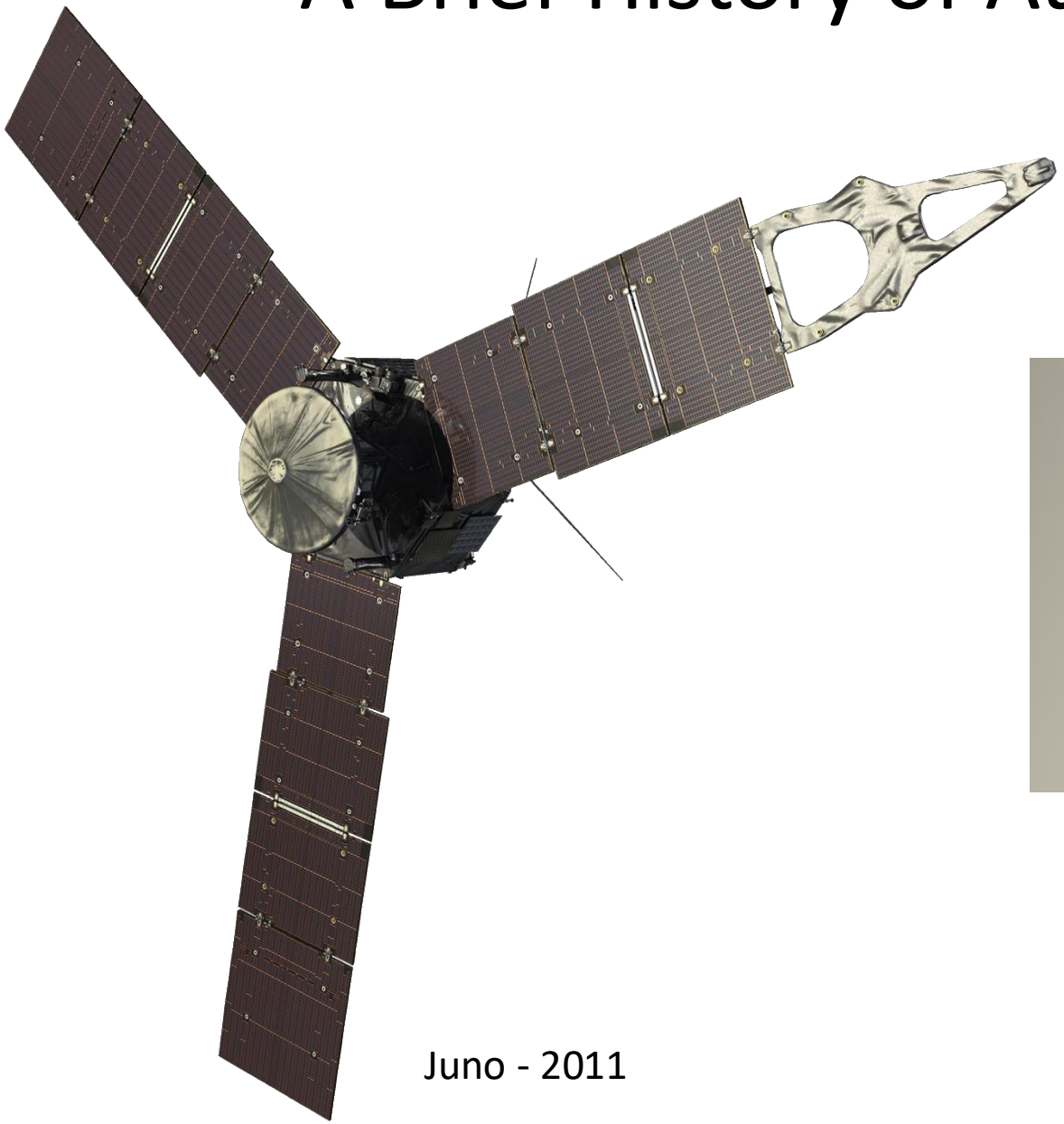


Discoverer 2 (Corona) - 1959

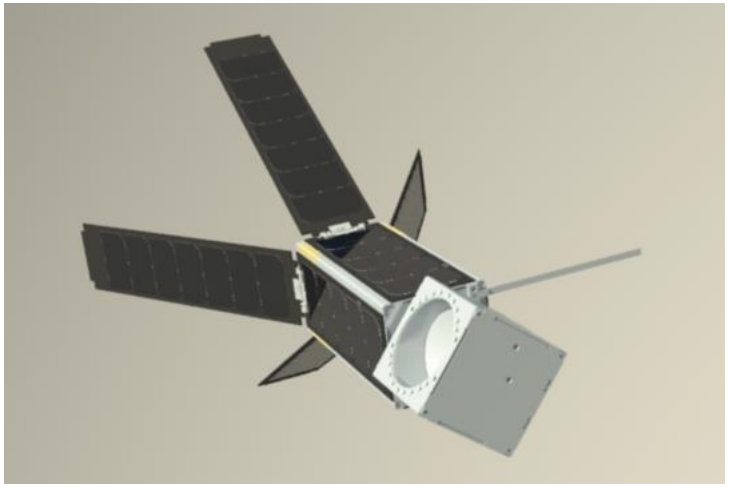
A Brief History of Attitude…

Intelsat 5 - 1980

A Brief History of Attitude…

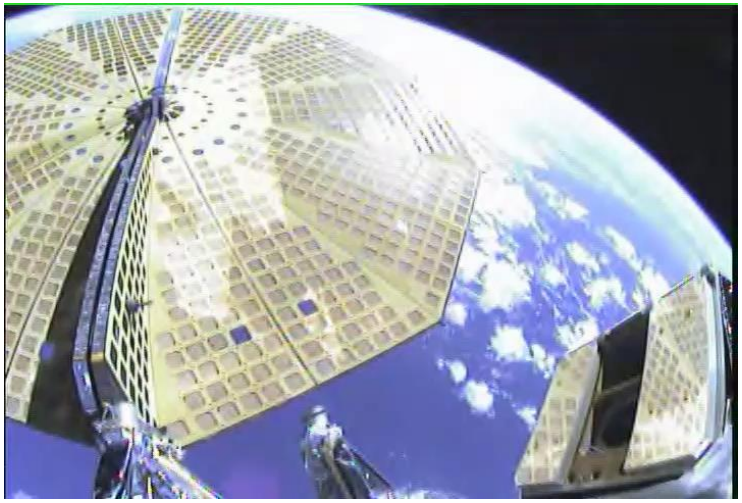


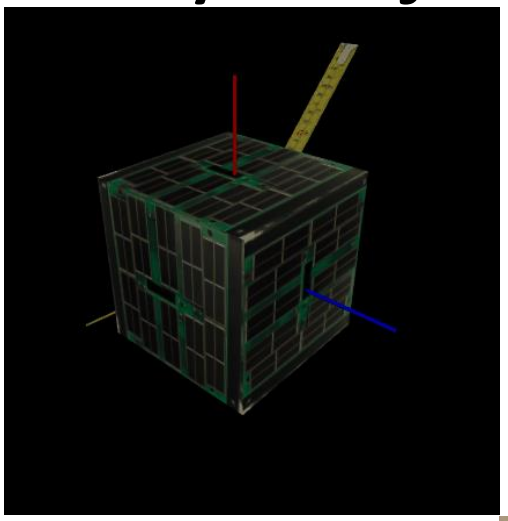
Juno - 2011

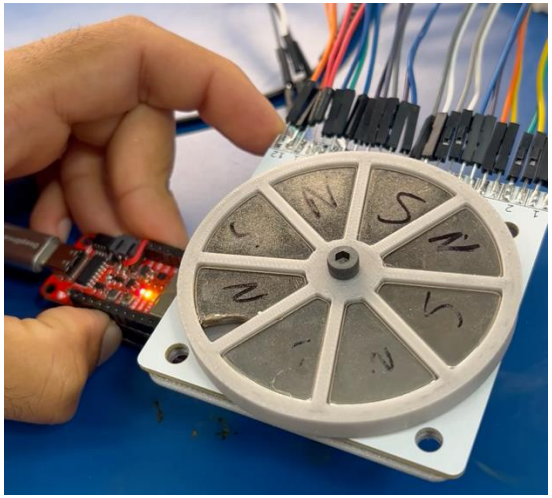


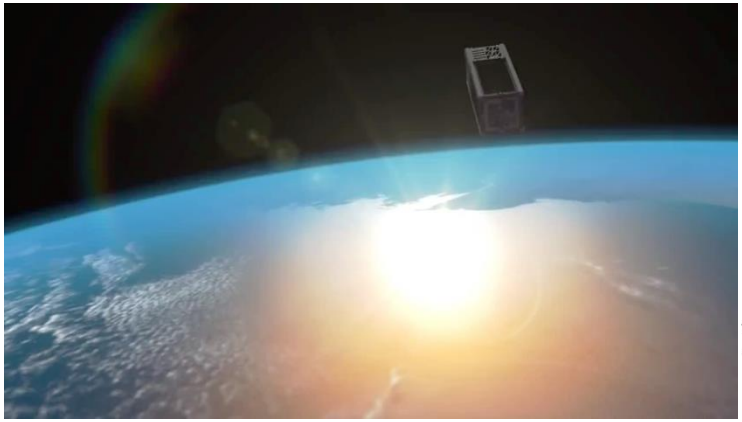
MicroMAS 2 - 2018

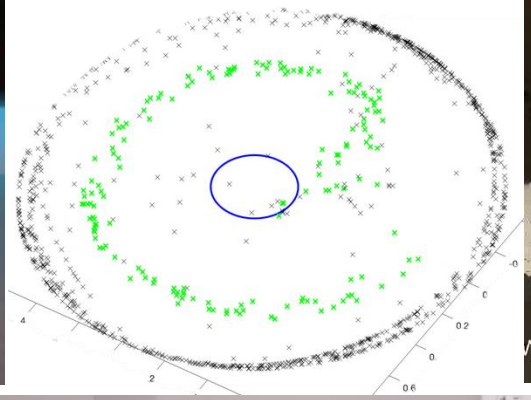
A Few of My Projects…

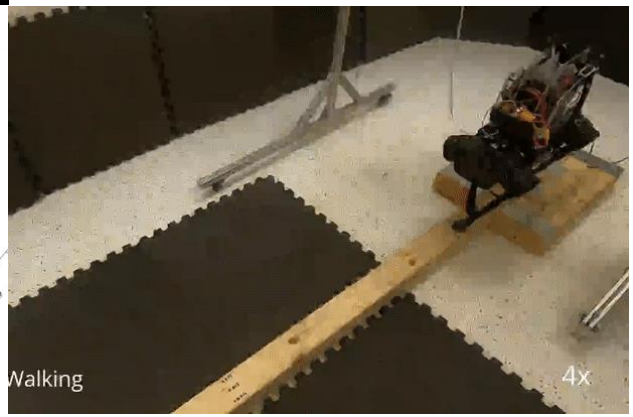


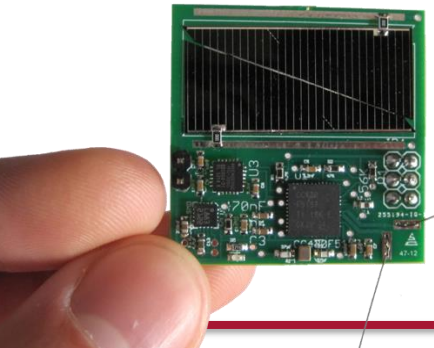












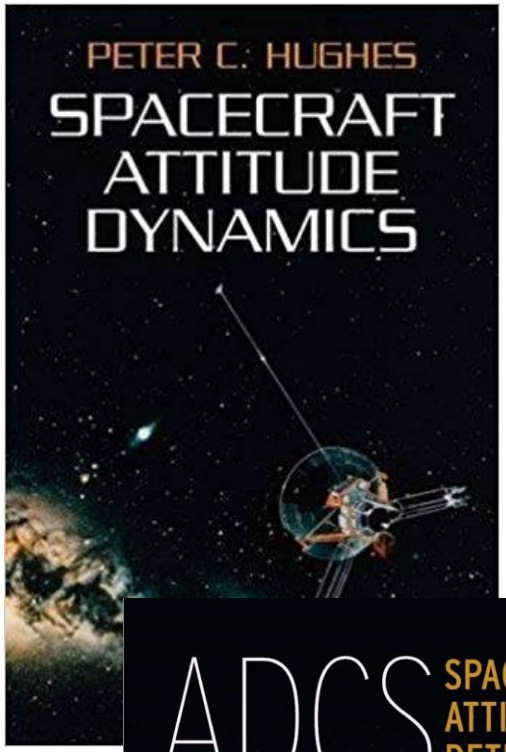
Rough Topic Schedule

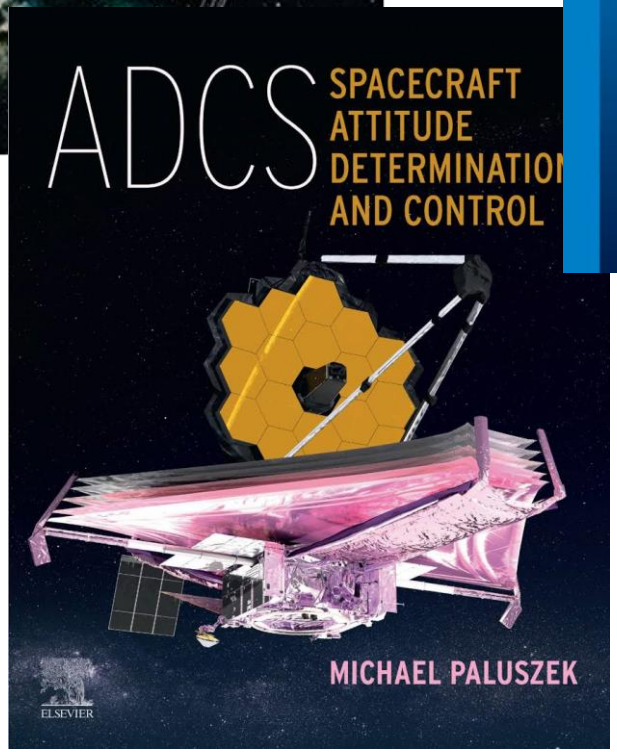
1. Attitude parameterizations and the SO(3) group
2. Rigid body and gyrostat dynamics
3. Damping and environmental perturbations
4. Spinning spacecraft and stability
5. Attitude determination: sensor measurements and TRIAD
6. Attitude determination: Whaba's problem and solution methods
7. Attitude determination: Kalman filters
8. Attitude control: passive solutions
9. Attitude control: feedback control
10. Calibration and Performance Analysis
11. Case Studies and Advanced Topics

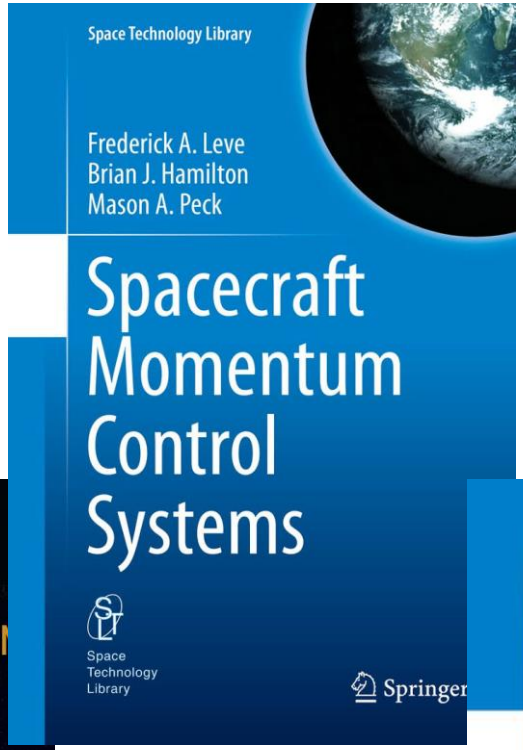
Grading and Project

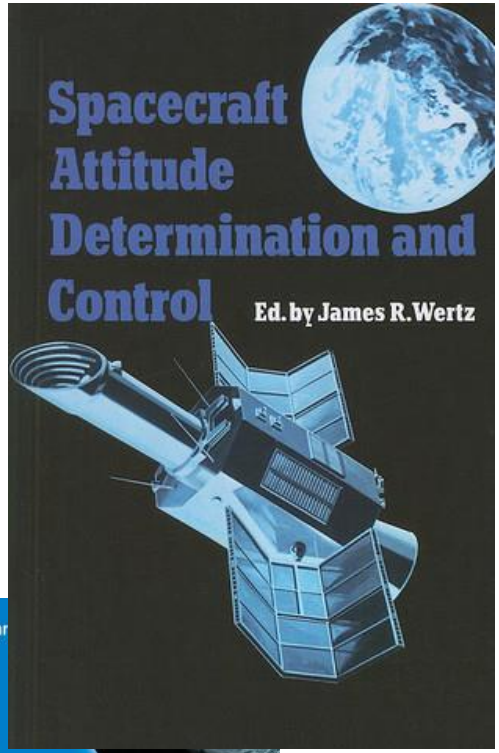
- Grading will be based on:
  - 40% homework
  - 40% final project
  - 15% quizzes
  - 5% participation
- Each homework will analyze some aspect of the ADCS design for a spacecraft of your choice
- I will give you feedback on each homework.
- The final project will be a compilation of the analysis from all of the homework assignments.

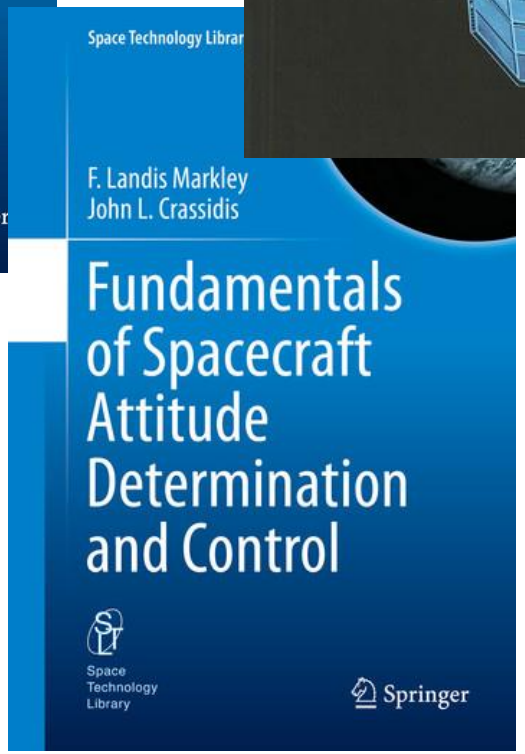
Books


PETER C. HUGHES
SPACECRAFT ATTITUDE DYNAMICS


ADCS
SPACECRAFT ATTITUDE DETERMINATION AND CONTROL
MICHAEL PALUSZEK
ELSEVIER


Space Technology Library
Frederick A. Leve
Brian J. Hamilton
Mason A. Peck
Spacecraft Momentum Control Systems
Space Technology Library
Springer


Spacecraft Attitude Determination and Control
Ed. by James R. Wertz


Space Technology Librar
F. Landis Markley
John L. Crassidis
Fundamentals of Spacecraft Attitude Determination and Control
Space Technology Library
Springer

“Assignment Zero”

Fill out course survey: https://forms.gle/eStyuzbNgjxQ6jLu7